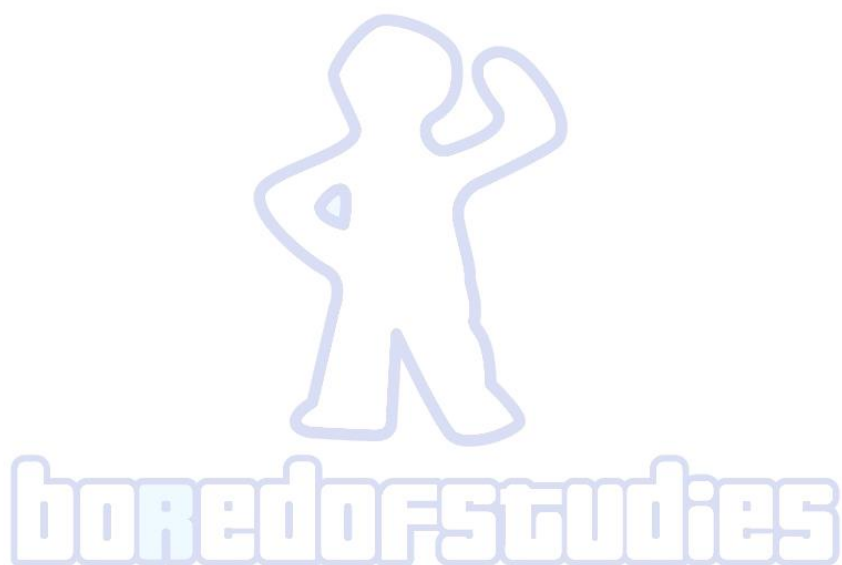




2016 Bored of Studies Trial Examinations

Mathematics Extension 2

3rd October 2016

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used.
- A reference sheet has been provided.
- Show all necessary working in Questions 11 – 16.

Total Marks – 100

Section I Pages 1 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 7 – 25

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

Total marks – 10

Attempt Questions 1 – 10

All questions are of equal value

Shade your answers in the appropriate box in the Multiple Choice answer sheet provided.

- 1 A particle of mass m is attached to a fixed point on a smooth table by an inextensible string of length r . It is then spun around in uniform circular motion with angular velocity ω . Let the tension that the string exerts be T .

The radius, angular velocity and mass of the particle is allowed to vary. Which of the following changes will result in a DIFFERENT value of T from the other options?

- (A) The mass is quadrupled, but the angular velocity is halved.
 - (B) The mass halved, but the angular velocity quadrupled.
 - (C) The radius is halved, but the mass is doubled.
 - (D) The mass is halved, and the radius is doubled.
- 2 Which of the following statements is always TRUE?
- (A) If $P(x)$ has real coefficients, and $x = a + bi$ is a root, then $x = a - bi$ is also a distinct root, for *all* real values of a and b .
 - (B) A cubic polynomial can *never* have exactly one non-real root.
 - (C) If a polynomial with real coefficients has n distinct non-real roots, then it *must* have degree $2n$.
 - (D) If a monic polynomial with integer coefficients has no integer roots, then its roots *must* be either irrational or non-real.

3 Consider the relation $e^y \sin x + e^x \sin y = 0$.

Which of the following is a correct expression for $\frac{dy}{dx}$?

(A) $e^{y-x} \left(\frac{\sin x - \cos x}{\cos y + \sin y} \right)$

(B) $e^{y-x} \left(\frac{\sin x + \cos x}{\cos y - \sin y} \right)$

(C) $e^{y-x} \left(\frac{\sin x - \cos x}{\cos y - \sin y} \right)$

(D) $e^{y-x} \left(\frac{\sin x + \cos x}{\cos y + \sin y} \right)$

4 Let $y = f(x)$ be any function that has an x intercept at $x = \alpha$.

Which of the following is ALWAYS true?

(A) The graph of $y = (f(x))^2$ will have a stationary point at $x = \alpha$.

(B) The graph of $y = (f(x))^3$ will have a horizontal point of inflexion at $x = \alpha$.

(C) The graph of $y = \sqrt{f(x)}$ will have a vertical tangent at $x = \alpha$.

(D) The graph of $y = \tan^{-1}(f(x))$ will be parallel to $y = f(x)$ at $x = \alpha$.

- 5 Let $a > 0$ be a fixed positive constant. Consider a continuous odd function that has been shifted upwards by a units. Let this shifted function be $y = f(x)$. The function must satisfy the following conditions

- $f(-a) = 0$
- $f(a) = 2a$
- $f(x) \geq 0$

Which of the following does not have a fixed value, for ANY $f(x)$ satisfying the above properties?

(A) $\int_{-a}^a f(x) dx.$

(B) $\int_{-a}^a f(x) + f(-x) dx.$

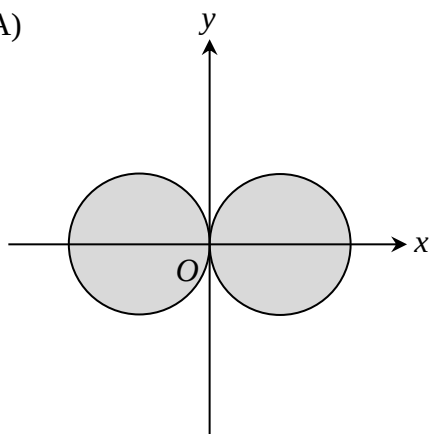
(C) $\int_0^a f(x) - f(a-x) dx.$

(D) $\int_{-a}^a f(x) + f(a-x) dx.$

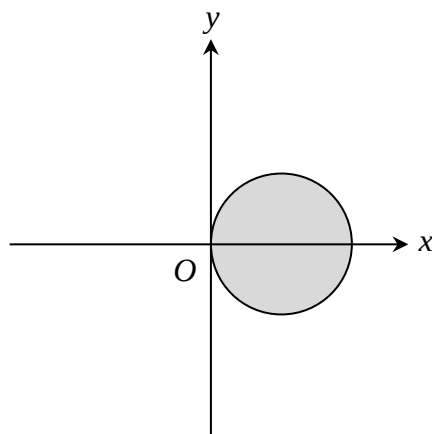
6 Which of the following diagrams best fits the locus equation described by

$$\left| \frac{1}{z} + \frac{1}{\bar{z}} \right| \geq 1$$

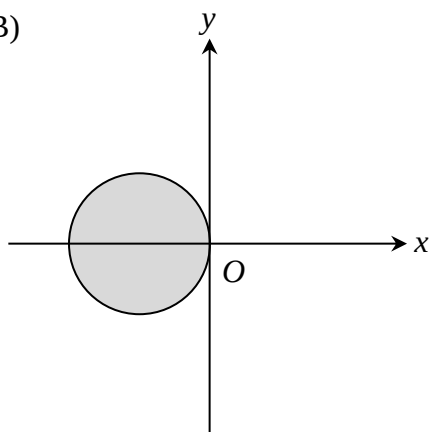
(A)



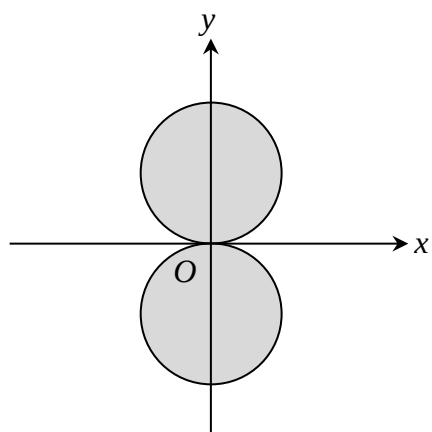
(C)



(B)



(D)



7 The integral

$$\int_{-2}^{-1} \frac{\sqrt{x^2-1}-2x^2\sqrt{3}}{x} dx,$$

is evaluated using the trigonometric substitution $x = \sec \theta$.

Which of the following is the correct value of the integral?

(A) $4\sqrt{3} - \frac{\pi}{3}$

(B) $4\sqrt{3} + \frac{\pi}{3}$

(C) $2\sqrt{3} - \frac{\pi}{3}$

(D) $2\sqrt{3} + \frac{\pi}{3}$

8 An ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, with $a > b > 0$, has one of its directrices being $x = 2$ and the minor axis length $2b$ is equal to k units.

For what values of k will such an ellipse exist?

(A) $0 < k \leq 2$.

(B) $-2 \leq k \leq 2$.

(C) $-1 \leq k \leq 1$.

(D) $0 < k \leq 1$.

- 9 A car of mass m moves along a banked track with speed v in a circle of radius r . The banking angle of the track is θ . Let g be the acceleration due to gravity.

The equation below is the expression for *friction* after forces have been resolved.

$$F = mg \sin \theta - \frac{mv^2}{r} \cos \theta, \text{ where } F > 0.$$

As v decreases, the value of F increases. When $v = v_F$, the magnitude of F attains a maximum $F_{\max}$, but the car will not slip. For all $0 < v < v_F$, the friction F maintains the same value $F_{\max}$ but the car will begin to slip sideways.

It is observed that at a particular moment, the car has some speed $0 < v < v_F$. Which of the following actions will best prevent the car from slipping further?

- (A) Decrease radius AND increase mass.
- (B) Decrease radius AND decrease mass.
- (C) Increase radius AND increase mass.
- (D) Increase radius AND decrease mass.

- 10 Let a , b and c be positive numbers. Which of the following expressions has the smallest minimum value?

- (A) $\frac{(a+b)(b+c)(a+c)}{abc}$.
- (B) $\frac{a+b}{c} + \frac{b+c}{a} + \frac{a+c}{b}$.
- (C) $\left(a + \frac{1}{a}\right)\left(b + \frac{1}{b}\right)\left(c + \frac{1}{c}\right)$.
- (D) $(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$.

Total marks – 90

Attempt Questions 11 – 16

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Suppose $f(x)$ is a non-linear polynomial where $f'(\alpha) = f(\alpha) = 0$ for a particular point $x = \alpha$.

(i) Show that $f(x)$ can be expressed in the form 2

$$f(x) = (x - \alpha)^2 g(x),$$

where $g(x)$ is some function that is continuous at $x = \alpha$.

(ii) Suppose $f(x)$ is not a polynomial. 1

By finding a counter-example, show that the result in (i) may no longer be true.

(b) (i) Sketch the graph of $y = \ln x - x$. 2

(ii) Hence, or otherwise, sketch the graph of $y = e^{x+y}$. 2

Question 11 continues on page 8

Question 11 (continued)

- (c) Every minute, the probability that a certain bacterium will divide into two identical bacteria is p . The probability that it will die is $1 - p$. A bacterium will either die or divide independently of the other bacteria in the population. Initially, the population consists of exactly one bacterium.

Let Q be the probability that the population eventually dies out.

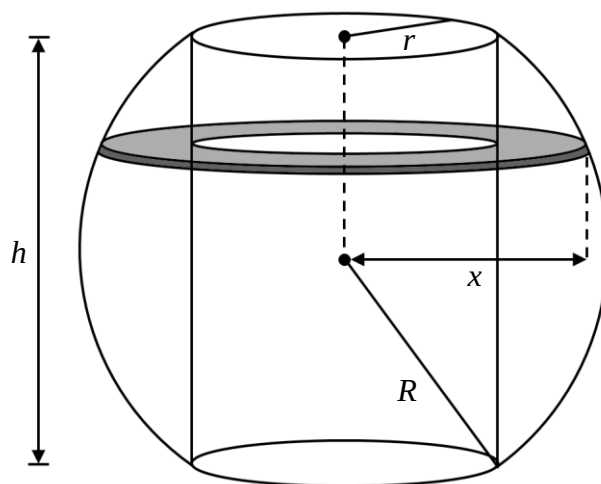
(i) Explain why $Q = (1 - p) + pQ^2$. 2

(ii) Hence show that if $0 \leq p \leq \frac{1}{2}$, then the population is guaranteed to eventually become zero. 2

Question 11 continues on page 9

Question 11 (continued)

- (d) The diagram below shows a sphere of radius R . A hole of height h and constant radius r is drilled through the centre of the sphere, leaving behind a solid.



A typical horizontal cross section with outer radius x is shaded.

- (i) Show that the volume of the solid is $\frac{\pi}{6}h^3$ units³. 3
- (ii) Consider two spheres with unequal radii. A hole of fixed height h is drilled through the centre of both spheres, forming solids $\mathcal{S}_1$ and $\mathcal{S}_2$. 1

What can be said about the volume of the two solids?

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) A particle of mass m is projected vertically upwards through a medium with initial velocity u . It experiences gravitational acceleration g and a resistive force mkv^2 , where k is a positive constant and v is the velocity of the particle at time t . The particle's displacement from the projection point, after t seconds, is given by x .

Let the terminal speed of the particle, when it falls back down through the same medium, be w .

- (i) Show that the terminal velocity is given by $w^2 = \frac{g}{k}$. 1

- (ii) Show that particle has velocity 3

$$v = w \left(\frac{u - w \tan(gt/w)}{w + u \tan(gt/w)} \right).$$

You may use the fact that if $0 < b < a$, then

$$\tan^{-1} a - \tan^{-1} b = \tan^{-1} \left(\frac{a - b}{1 + ab} \right).$$

without proof.

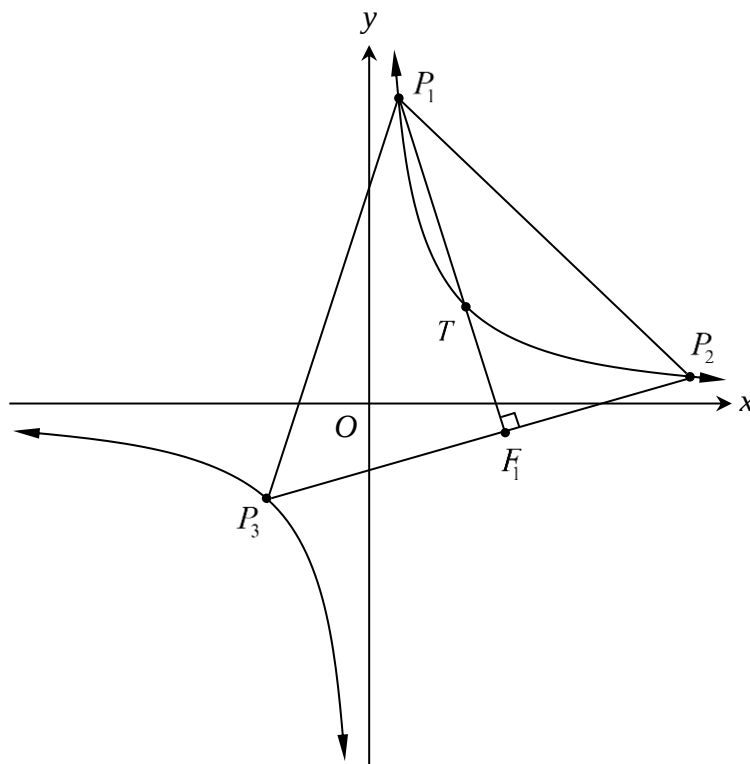
- (iii) Show that particle has displacement 2

$$x = \frac{w^2}{g} \ln \left(\frac{w \cos(gt/w) + u \sin(gt/w)}{w} \right).$$

Question 12 continues on page 11

Question 12 (continued)

- (b) Let $\left(ct, \frac{c}{t}\right)$, where $c > 0$, be any point on the hyperbola $xy = c^2$. The diagram below shows three points P_1 , P_2 and P_3 with parameters t_1 , t_2 , t_3 respectively. The line F_1P_1 is perpendicular to P_2P_3 and intersects the hyperbola at T .



- (i) Show that the equation of F_1P_1 is 2

$$t_1 t_2 t_3 x - t_1 y = c(t_1^2 t_2 t_3 - 1).$$

- (ii) Show that T has coordinates 2

$$\left(\frac{-c}{t_1 t_2 t_3}, -c t_1 t_2 t_3\right).$$

Question 12 continues on page 12

Question 12 (continued)

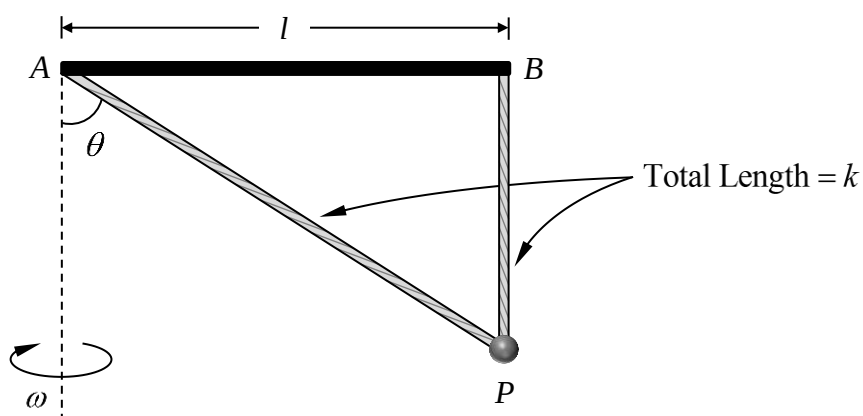
- (iii) Let F_2 and F_3 be the feet of the perpendiculars from P_2 and P_3 to P_1P_3 and P_1P_2 respectively. 2

Deduce that F_1P_1 , F_2P_2 and F_3P_3 are concurrent at T .

Question 12 continues on page 13

Question 12 (continued)

- (c) The diagram shows a straight rod AB of length l rotating in a horizontal plane about A . A light inextensible string of length k is attached to A and B . It is threaded through the centre of a bead P with mass m , which is free to slide smoothly along the string. When the angular speed of AB is ω , the bead remains vertically below B , and AP makes an acute angle θ with the vertical axis. Let g be the acceleration due to gravity.

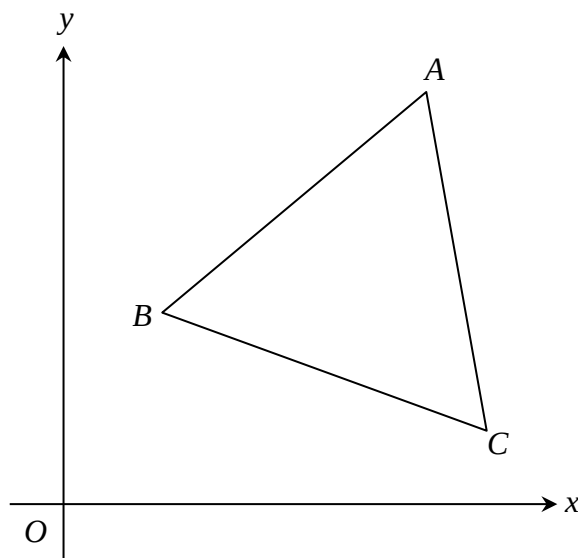


- (i) Show that $\tan\left(\frac{\theta}{2}\right) = \frac{l\omega^2}{g}$. 2
- (ii) Hence prove that $\omega^2 = \frac{g}{k}$. 1

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) Let a , b and c be complex numbers representing the vertices of $\triangle ABC$, labelled anti-clockwise in the Argand diagram. Let $\omega = \cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)$.



- (i) Suppose $\triangle ABC$ is equilateral. 2

Show that $a + b\omega + c\omega^2 = 0$.

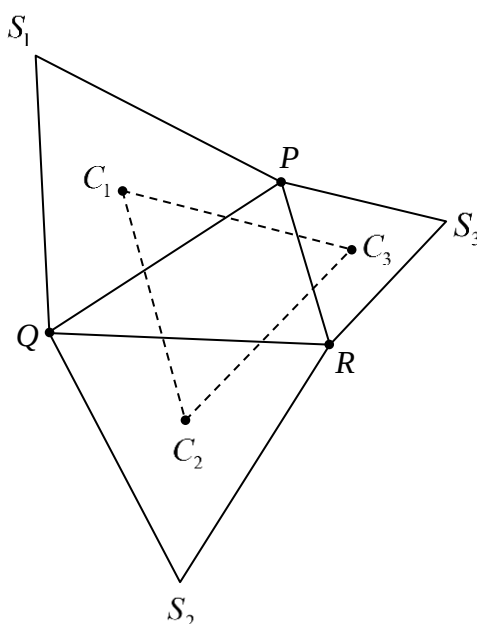
- (ii) Show that conversely, if $a + b\omega + c\omega^2 = 0$, then $\triangle ABC$ is an equilateral triangle. 2

Question 13 continues on page 15

Question 13 (continued)

- (b) Let p, q and r be complex numbers representing the vertices of $\triangle PQR$. 2

Each side of $\triangle PQR$ is one of the sides of an equilateral triangle with the third vertex being S_1, S_2 and S_3 , as shown in the diagram below, represented by complex numbers s_1, s_2 and s_3 .



It can be shown that the centre C_1 of $\triangle PQS_1$ is represented by the complex number

$$c_1 = \frac{p + q + s_1}{3},$$

and similar results hold for c_2 and c_3 corresponding to C_2 and C_3 respectively.

Use part (a) to prove that $\triangle C_1C_2C_3$ is equilateral.

Question 13 continues on page 16

Question 13 (continued)

(c) Let $P(x)$ be a monic polynomial of degree n with real distinct roots α_k , where

$k = 1, 2, 3, \dots, n$. Using partial fractions, $\frac{1}{P(x)}$ can be expressed in the form

$$\frac{1}{P(x)} = \sum_{k=1}^n \frac{c_k}{x - \alpha_k},$$

for some real c_k .

(i) Show that

1

$$c_1 = \frac{x - \alpha_1}{P(x)} - \sum_{k=2}^n c_k \left(\frac{x - \alpha_1}{x - \alpha_k} \right)$$

for some real c_k .

(ii) Hence show that

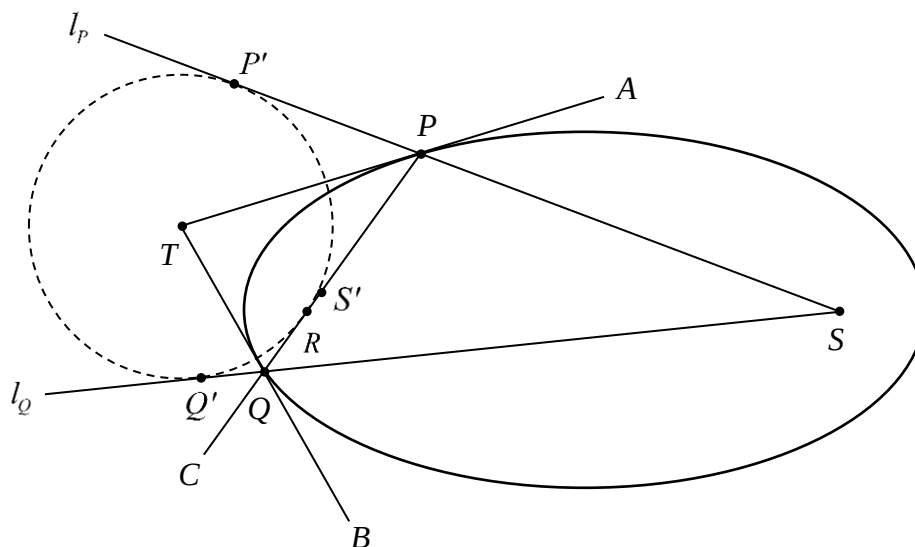
2

$$\frac{1}{P(x)} = \sum_{k=1}^n \frac{1}{P'(\alpha_k)(x - \alpha_k)}.$$

Question 13 continues on page 17

Question 13 (continued)

- (d) The diagram below shows a focal chord PQ on an arbitrary ellipse with foci S and S' . The chords SP and SQ are produced to form lines l_p and l_q respectively. The tangents drawn from P and Q intersect at T .



- (i) Show that T is the centre of a circle $\mathcal{C}$ that is tangential to l_p , l_q and PQ at P' , Q' and R respectively. 3
- (ii) Let R be the point of contact between the chord PQ and $\mathcal{C}$.
Show that $PR + PS = QR + QS$. 1
- (iii) Deduce that R and S' coincide. 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) Let x be a positive real number and $n \geq 2$ be an integer.

(i) Use mathematical induction to prove that **3**

$$(1+x)^n \geq 1+nx.$$

(ii) Hence show that if $x > \frac{n}{n-1}$, then **2**

$$x^n > x+1.$$

(b) Let $P(z) = z^n + z + 1$ be a non-linear polynomial. **2**

Use part (a) to prove that the complex roots of $P(z)$ lie in the region

$$|z| \leq \frac{n}{n-1}.$$

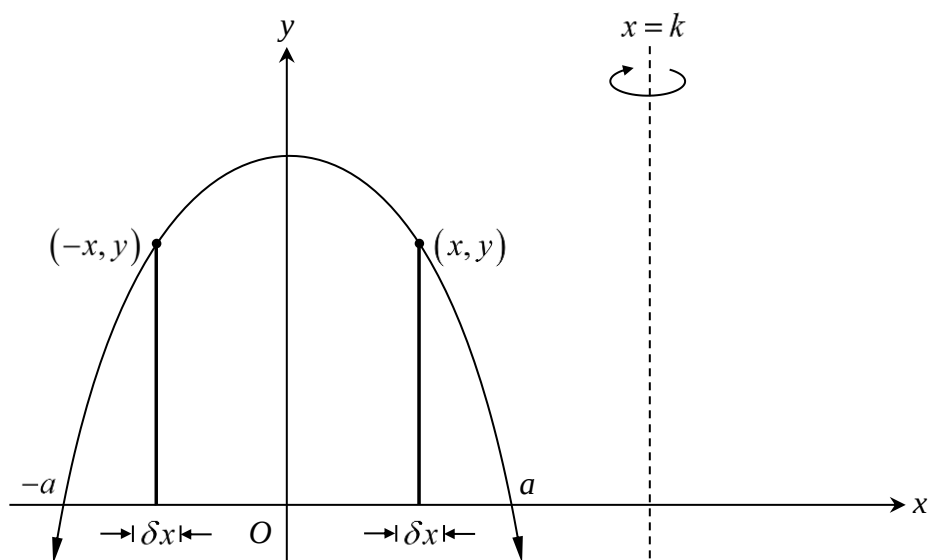
Question 14 continues on page 19

Question 14 (continued)

- (c) Let $y = f(x)$ be an even positive function in the domain $-a \leq x \leq a$ with $f(a) = f(-a) = 0$.

The region bounded by the curve and the x axis is rotated about the line $x = k$, forming a solid with volume V with cross sectional area A .

The diagram below shows two typical vertical strips of width δx at (x, y) and $(-x, y)$, which form shells with volume δV_1 and δV_2 respectively when rotated about $x = k$.



- (i) Show that $\delta V_2 \approx 2\pi f(x)(k+x)\delta x$ and state a similar result for δV_1 . 2
- (ii) Deduce that the ratio between V and A is independent of the function. 1

Question 14 continues on page 20

Question 14 (continued)

(d) Consider the integral

$$I_n = \int_0^1 \frac{x^{4n} (1-x)^{4n}}{1+x^2} dx.$$

where n is a positive integer.

(i) Show that 2

$$\frac{x^{4n} (1-x)^{4n}}{1+x^2} = P(x) + \frac{2^{2n} (-1)^n}{1+x^2},$$

where $P(x)$ is a polynomial of degree $8n-2$.

(ii) Show that $I_n \rightarrow 0$ as $n \rightarrow \infty$. 1

(iii) Show that 1

$$I_n = 2^{2n-2} (-1)^n \pi + R_n,$$

where R_n is some rational number.

(iv) Hence, explain the significance of the number $\left| \frac{R_n}{2^{2n}} \right|$ for large values 1
of n .

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) Prove that for $0 < x < \frac{\pi}{2}$, 3

$$\cot x < \frac{1}{x} < \operatorname{cosec} x.$$

- (b) Consider the equation

$$\left(\frac{z+i}{z-i} \right)^{2n} = -1,$$

where n is a positive integer.

- (i) Show that the roots are $z = \cot \left(\frac{(2k-1)\pi}{4n} \right)$ where $k = 1, 2, \dots, 2n$. 2

- (ii) Hence, or otherwise, find the roots of the polynomial 2

$$P(x) = \binom{2n}{0} x^n - \binom{2n}{2} x^{n-1} + \binom{2n}{4} x^{n-2} + \dots + (-1)^n \binom{2n}{2n}.$$

- (iii) Deduce that 2

$$\operatorname{cosec}^2 \left(\frac{\pi}{4n} \right) + \operatorname{cosec}^2 \left(\frac{3\pi}{4n} \right) + \operatorname{cosec}^2 \left(\frac{5\pi}{4n} \right) + \dots + \operatorname{cosec}^2 \left(\frac{(2n-1)\pi}{4n} \right) = 2n^2.$$

Question 15 continues on page 22

Question 15 (continued)

- (iv) Use the results in part (a) to deduce that 2

$$\frac{\pi^2}{16} \left(\frac{2n-1}{n} \right) < \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots + \frac{1}{(2n-1)^2} < \frac{\pi^2}{8}.$$

- (v) Hence, find the limiting value of 1

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

as n gets large.

- (c) (i) By considering a suitable binomial expansion, show that 2

$$2^n \cos^n \theta \cos n\theta = \sum_{k=0}^n \binom{n}{k} \cos 2k\theta.$$

- (ii) Hence, find 1

$$\int_{-\pi}^{\pi} \cos^n \theta \cos n\theta d\theta.$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) A bag has m white balls and $n - m$ black balls. An integer k from 0 to n inclusive is chosen randomly, and then k balls are drawn from the bag without replacement. 2

Find the probability that all drawn balls are white.

- (b) Suppose there are m white balls, $n - m$ black balls and one red ball arranged randomly in a row from left to right.

- (i) Explain why the probability that the red ball is to the left of all the black balls is $\frac{1}{n - m + 1}$. 1

- (ii) Use part (a), or otherwise, to simplify 2

$$\frac{\binom{m}{0}}{\binom{n}{0}} + \frac{\binom{m}{1}}{\binom{n}{1}} + \frac{\binom{m}{2}}{\binom{n}{2}} + \dots + \frac{\binom{m}{m}}{\binom{n}{m}}.$$

Question 16 continues on page 24

Question 16 (continued)

- (c) Let r be a rational number $\frac{a}{b}$ where a and b are non-zero integers. Suppose e^r can be expressed in the form $\frac{p}{q}$, where p and q are integers.

Define the family of integrals

$$I_n = \frac{a^{2n+1}pq}{n!} \int_{-1}^1 (1-x^2)^n e^{rx} dx,$$

for $n = 0, 1, 2, 3, \dots$

You are given that $I_1 = 2ab^2(p^2 + q^2) - 2b^3(p^2 - q^2)$. (Do NOT prove this.)

- (i) Show that

4

$$I_n = 4a^2b^2I_{n-2} - 2(2n-1)b^2I_{n-1},$$

for all integers $n \geq 2$.

- (ii) Prove that

2

$$0 < |I_n| \leq 2pq \frac{|a|^{2n+1}}{n!} e^{|r|}.$$

Question 16 continues on page 25

Question 16 (continued)

- (iii) It can be shown that if R is any real number, then as $n \rightarrow \infty$, **3**

$$\frac{R^n}{n!} \rightarrow 0.$$

Use this to prove that if r is any non-zero rational number, then e^r is irrational.

- (iv) Deduce that if r is a positive rational number other than 1, then **1**
 $\log_e(r)$ is also irrational.

End of Exam